

FAKE JOB POSTONG CLASSIFIER

Project Description

This project detects whether a job posting is real or fake using Machine Learning. It analyzes the job title, company profile, description, requirements, and benefits to classify job advertisements. Text data is converted into numerical features using TF-IDF, and a Random Forest Classifier is trained to make predictions

0 = Real Job 1 = Fake Job

Model Used

Random Forest Classifier

The model used in this project is the Random Forest Classifier, which is an ensemble machine learning algorithm.

It consists of multiple decision trees, and each tree makes its own prediction. The final prediction is determined by majority voting among all the trees

IMPORTING LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from wordcloud import WordCloud
import joblib
```

LOADING DATASET

```
with open("fake_job_postings.csv", "r", errors="ignore") as f:
    for i in range(10):
        print(f.readline())
```

```
job_id,title,location,department,salary_range,company_profile,description,requirements,benefits,telecommuting,has_company_logo,h
1,Marketing Intern,"US, NY, New York",Marketing,,,"We're Food52, and we've created a groundbreaking and award-winning cooking sit
2,Customer Service - Cloud Video Production,"NZ, , Auckland",Success,,,"90 Seconds, the worlds Cloud Video Production Service.90
3,Commissioning Machinery Assistant (CMA),"US, IA, Wever",,,,"Valor Services provides Workforce Solutions that meet the needs of
4,Account Executive - Washington DC,"US, DC, Washington",Sales,,,"Our passion for improving quality of life through geography is
5,Bill Review Manager,"US, FL, Fort Worth",,,,"SpotSource Solutions LLC is a Global Human Capital Management Consulting firm head
6,Accounting Clerk,"US, MD, ",,,,"Job OverviewApex is an environmental consulting firm that offers stable leadership and growth
7,Head of Content (m/f),"DE, BE, Berlin",ANDROIDPIT,20000-28000,"Founded in 2009, the Fonpit AG rose with its international web
8,Lead Guest Service Specialist ,,"US, CA, San Francisco",,,,"Airenvys mission is to provide lucrative yet hassle free full serv
9,HP BSM SME,"US, FL, Pensacola",,,,"Solutions3 is a woman-owned small business whose focus is IT Service Management using best o
```

```
import os

print("File size:", os.path.getsize("fake_job_postings.csv"), "bytes")
```

File size: 6291456 bytes

```
import pandas as pd
```

```
df = pd.read_csv(
    "fake_job_postings.csv",
    header=None,
    on_bad_lines="skip",
    engine="python"
)

df.head()
```

	0	1	2	3	4	5	6	7	8	
	job_id	title	location	department	salary_range	company_profile	description	requirements	benefits	telecomm
0	1	Marketing Intern	US, NY, New York	Marketing	NaN	We're Food52, and we've created a groundbreaki...	Food52, a fast-growing, James Beard Award-winn...	Experience with content management systems a m...	NaN	
1	2	Customer Service - Cloud Video Production	NZ , Auckland	Success	NaN	90 Seconds, the worlds Cloud Video Production ...	Organised - Focused - Vibrant - Awesome!Do you...	What we expect from you:Your key responsibilit...	What you will get from usThrough being part of...	
2	3	Commissioning Machinery Assistant (CMA)	US, IA, Wever	NaN	NaN	Valor Services provides Workforce Solutions th...	Our client, located in Houston, is actively se...	Implement pre-commissioning and commissioning ...	NaN	
3	4	Account Executive - Washington DC	US, DC, Washington	Sales	NaN	Our passion for improving quality of life thro...	THE COMPANY: ESRI – Environmental Systems Rese...	EDUCATION: Bachelor's or Master's in GIS, busi...	Our culture is anything but corporate —we have ...	

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
print(df.shape)
```

(2644, 18)

```
print(df.columns)
```

Index([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17], dtype='int64')

```
df.head()
```

	7	8
requirements		benefits telecomm
with content systems a m...		NaN
expect from J:Your key possibilit...		What you will get from usThrough being part of...
ement pre- ioning and ssioning ...		NaN
Bachelor's r's in GIS, busi...		Our culture is anything but corporate —we have ...

```
df.columns = df.iloc[0]
df = df[1:].reset_index(drop=True)

df.head()
```

job_id		title	location	department	salary_range	company_profile	description		requirements	benefits
0	1	Marketing Intern	US, NY, New York	Marketing	NaN	We're Food52, and we've created a groundbreaki...	Food52, a fast-growing, James Beard Award-winn...		Experience with content management systems a m...	N
1	2	Customer Service - Cloud Video Production	NZ, , Auckland	Success	NaN	90 Seconds, the worlds Cloud Video Production ...	Organised - Focused - Vibrant - Awesome!Do you...		What we expect from you:Your key responsibilit...	What y will fr usThrou being p c
2	3	Commissioning Machinery Assistant (CMA)	US, IA, Wever	NaN	NaN	Valor Services provides Workforce Solutions th...	Our client, located in Houston, is actively se...		Implement pre-commissioning and commissioning ...	N
3	4	Account Executive - Washington DC	US, DC, Washington	Sales	NaN	Our passion for improving quality of life thro...	THE COMPANY: ESRI – Environmental Systems Rese...		EDUCATION: Bachelor's or Master's in GIS, busi...	cultur anyth corpor—we ha
4	5	Bill Review Manager	US, FL, Fort Worth	NaN	NaN	SpotSource Solutions LLC is a Global Human Cap...	JOB TITLE: Itemization Review ManagerLOCATION:...		QUALIFICATIONS:RN license in the State of Texa...	I Bene Offe

```
print(df.columns)

Index(['job_id', 'title', 'location', 'department', 'salary_range',
       'company_profile', 'description', 'requirements', 'benefits',
       'telecommuting', 'has_company_logo', 'has_questions', 'employment_type',
       'required_experience', 'required_education', 'industry', 'function',
       'fraudulent'],
      dtype='object', name=0)
```

Missing Values

```
df.isnull().sum()
```

	0
job_id	0
title	0
location	38
department	1670
salary_range	2225
company_profile	534
description	0
requirements	378
benefits	1060
telecommuting	0
has_company_logo	0
has_questions	0
employment_type	542
required_experience	1003
required_education	1100
industry	708
function	920
fraudulent	0

dtype: int64

Fill Missing Values

```
df=df.fillna("")
```

Check Fake vs Real Jobs

```
df['fraudulent'].value_counts()
```

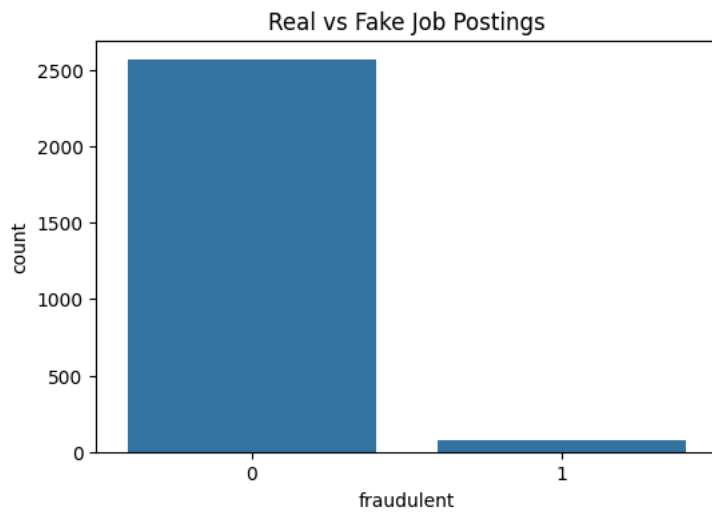
	count
fraudulent	
0	2564
1	79

dtype: int64

Plot Fake vs Real Jobs

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(6,4))
sns.countplot(x='fraudulent', data=df)
plt.title("Real vs Fake Job Postings")
plt.show()
```



Check Dataset Information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2643 entries, 0 to 2642
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   job_id                2643 non-null   object
1   title                 2643 non-null   object
2   location              2643 non-null   object
3   department            2643 non-null   object
4   salary_range          2643 non-null   object
5   company_profile       2643 non-null   object
6   description           2643 non-null   object
7   requirements          2643 non-null   object
8   benefits              2643 non-null   object
9   telecommuting         2643 non-null   object
10  has_company_logo      2643 non-null   object
11  has_questions         2643 non-null   object
12  employment_type       2643 non-null   object
13  required_experience    2643 non-null   object
14  required_education    2643 non-null   object
15  industry              2643 non-null   object
16  function              2643 non-null   object
17  fraudulent            2643 non-null   object
dtypes: object(18)
memory usage: 371.8+ KB
```

Missing Values

```
df.isnull().sum()
```

```

      0
      0
  job_id    0
    title    0
    location  0
  department  0
  salary_range  0
  company_profile  0
    description  0
    requirements  0
      benefits  0
  telecommuting  0
  has_company_logo  0
    has_questions  0
  employment_type  0
  required_experience  0
  required_education  0
      industry  0
      function  0
    fraudulent  0

```

dtype: int64

Fill Missing Values

```
df=df.fillna(" ")
```

Check Target Classes

```
df['fraudulent'].value_counts()
```

```

      count
fraudulent
0         2564
1          79

```

dtype: int64

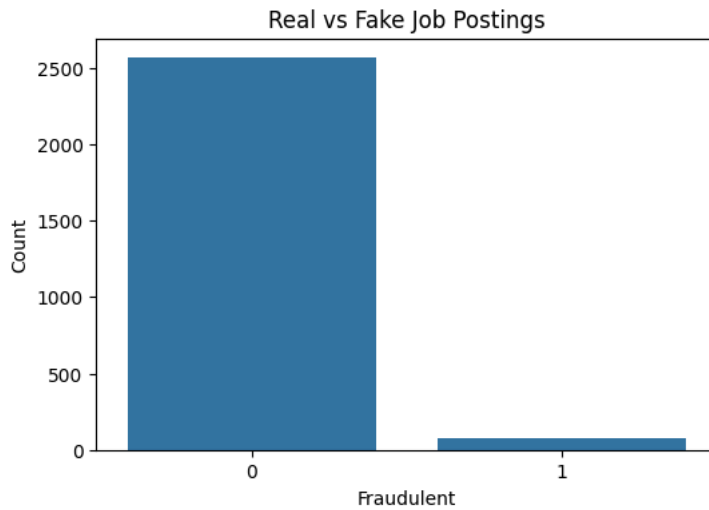
Plot Real vs Fake Jobs

```

import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(6,4))
sns.countplot(x='fraudulent', data=df)
plt.title("Real vs Fake Job Postings")
plt.xlabel("Fraudulent")
plt.ylabel("Count")
plt.show()

```



Combine Text Columns

```
df['text'] = (
    df['title'].astype(str) + " " +
    df['company_profile'].astype(str) + " " +
    df['description'].astype(str) + " " +
    df['requirements'].astype(str) + " " +
    df['benefits'].astype(str)
)
```

Word Cloud

```
!pip install wordcloud
```

```
Requirement already satisfied: wordcloud in /usr/local/lib/python3.12/dist-packages (1.9.6)
Requirement already satisfied: numpy>=1.19.3 in /usr/local/lib/python3.12/dist-packages (from wordcloud) (2.0.2)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from wordcloud) (11.3.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from wordcloud) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (26.2)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (2.9)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil->matplotlib->wordcloud) (1.16.0)
```

```
from wordcloud import WordCloud

text = " ".join(df['text'])

wordcloud = WordCloud(
    width=900,
    height=500,
    background_color='white'
).generate(text)

plt.figure(figsize=(15,7))
plt.imshow(wordcloud)
plt.axis("off")
plt.show()
```



Features and Labels

```
X = df['text']
y = df['fraudulent']
```

Train/Test Split

```
from sklearn.model_selection import train_test_split

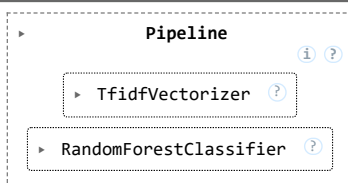
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

Build and Train Model

```
from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.ensemble import RandomForestClassifier

model = Pipeline([
    ('tfidf', TfidfVectorizer(stop_words='english', max_features=10000)),
    ('rf', RandomForestClassifier(n_estimators=200, random_state=42))
])

model.fit(X_train, y_train)
```



Predict on Test Data

```
prediction = model.predict(X_test)
```

Accuracy

```
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, prediction)

print("Accuracy:", accuracy)
```

Accuracy: 0.9924385633270322

Classification Report

```
from sklearn.metrics import classification_report

print(classification_report(y_test, prediction))
```

	precision	recall	f1-score	support
0	0.99	1.00	1.00	512
1	1.00	0.76	0.87	17
accuracy			0.99	529
macro avg	1.00	0.88	0.93	529
weighted avg	0.99	0.99	0.99	529

Confusion Matrix

```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, prediction)

print(cm)
```

```
[[512  0]
 [ 4  13]]
```

Confusion Matrix Graph

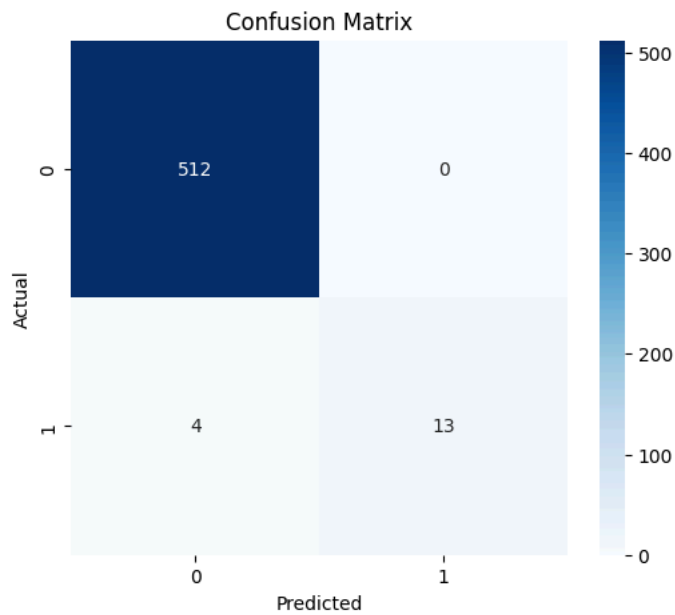
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(6,5))

sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Blues')

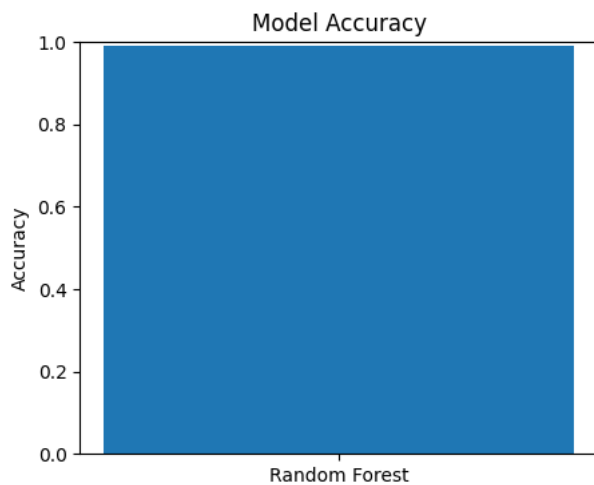
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.show()
```



Accuracy Bar Graph

```
plt.figure(figsize=(5,4))  
plt.bar(["Random Forest"], [accuracy])  
plt.ylim(0,1)  
plt.title("Model Accuracy")  
plt.ylabel("Accuracy")  
plt.show()
```



Save Model

```
import joblib  
joblib.dump(model, "FakeJobModel.pkl")  
print("Model Saved Successfully")
```

Model Saved Successfully

Download Model

```
from google.colab import files  
  
files.download("FakeJobModel.pkl")
```

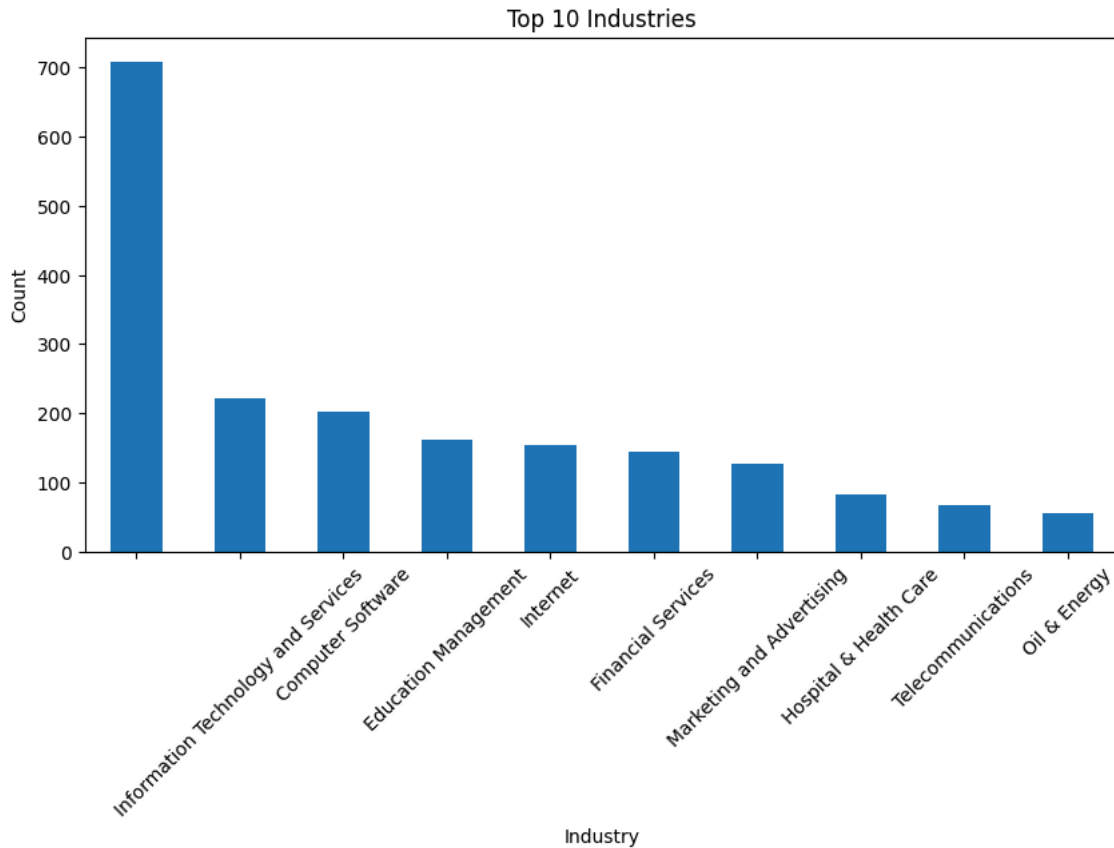
Test with a New Job Posting

```
sample = ""  
Urgently Hiring AI Engineer  
Salary: $120000  
Remote Job  
Medical Insurance  
Bachelor Degree Required  
Python, Machine Learning, Deep Learning  
""  
result = model.predict([sample])  
if result[0] == '1' or result[0] == 1:  
    print(" Fake Job Posting")  
else:  
    print("Real Job Posting")
```

Real Job Posting

Top 10 Industries

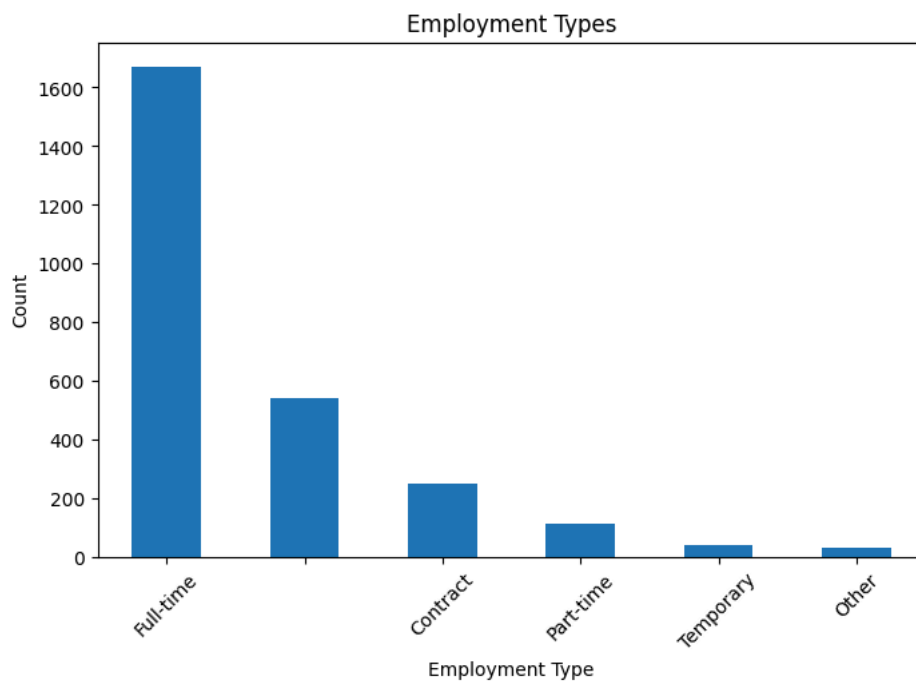
```
plt.figure(figsize=(10,5))  
df['industry'].value_counts().head(10).plot(kind='bar')  
plt.title("Top 10 Industries")  
plt.xlabel("Industry")  
plt.ylabel("Count")  
plt.xticks(rotation=45)  
plt.show()
```



Employment Types

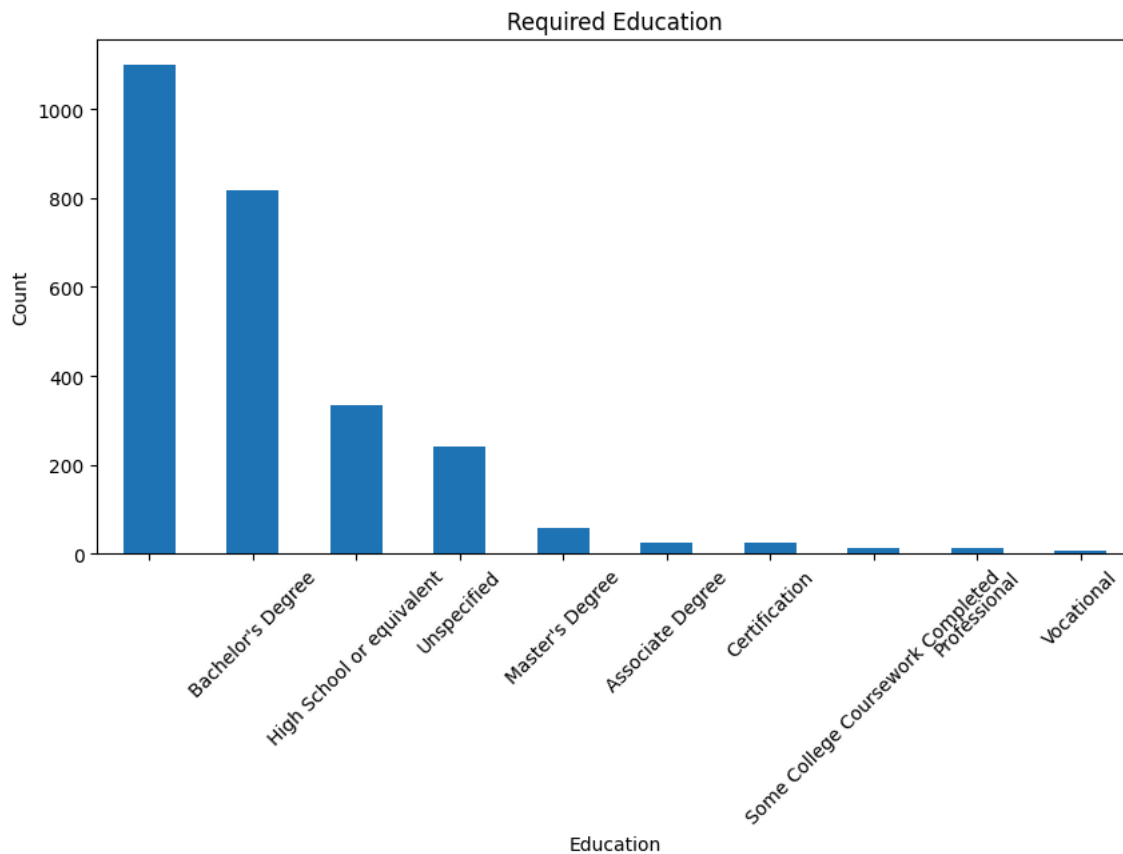
```
plt.figure(figsize=(8,5))  
df['employment_type'].value_counts().plot(kind='bar')  
plt.title("Employment Types")
```

```
plt.xlabel("Employment Type")  
plt.ylabel("Count")  
plt.xticks(rotation=45)  
plt.show()
```



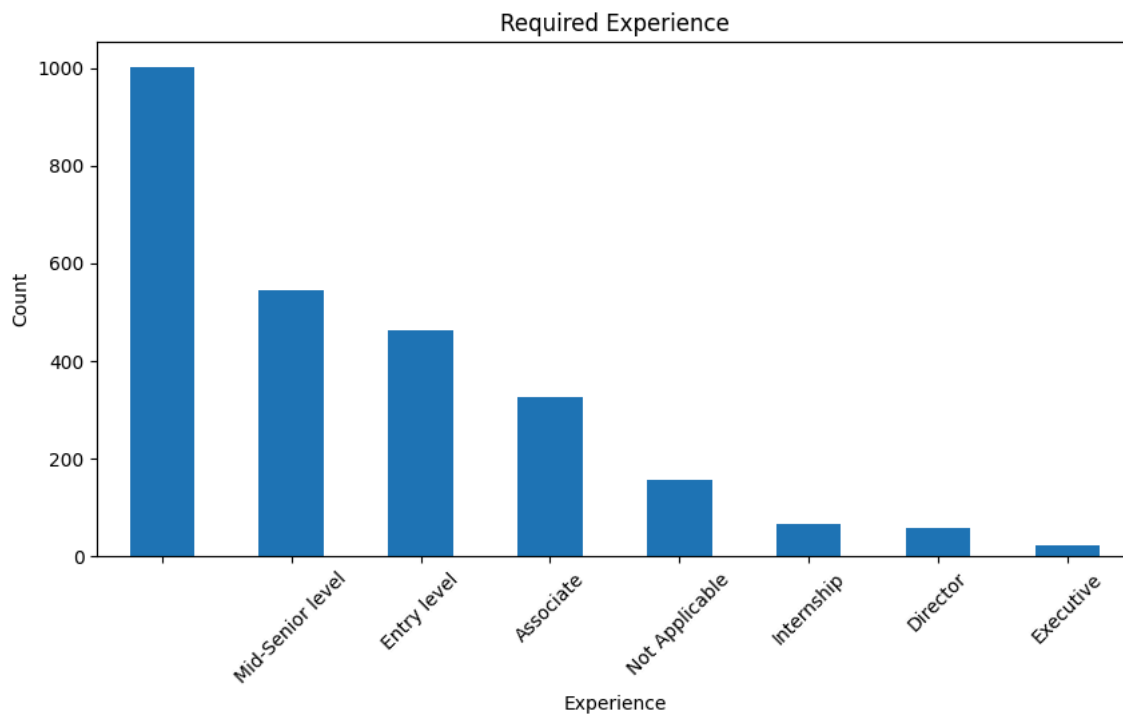
Required Education

```
plt.figure(figsize=(10,5))  
df['required_education'].value_counts().head(10).plot(kind='bar')  
plt.title("Required Education")  
plt.xlabel("Education")  
plt.ylabel("Count")  
plt.xticks(rotation=45)  
plt.show()
```



Required Experience

```
plt.figure(figsize=(10,5))
df['required_experience'].value_counts().head(10).plot(kind='bar')
plt.title("Required Experience")
plt.xlabel("Experience")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
```



Top Hiring Locations

```
plt.figure(figsize=(10,5))
df['location'].value_counts().head(10).plot(kind='bar')
plt.title("Top Job Locations")
plt.xlabel("Location")
plt.ylabel("Jobs")
plt.xticks(rotation=45)
plt.show()
```